Pavel Urysohn — Zum Metrisationsproblem (1925)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 17 October 2026.

Disclosure: prepared with AI assistance — the six subject reviewers named in the index are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

To Pavel Samuilovich Urysohn, who had three years in topology and left it the theory of dimension, a definition of compactness, and two theorems the subject still stands on — and to Paul Alexandroff, who edited the second paper after Urysohn's death and said so in a signed footnote.

The author's own terms

"The Theorem" (§25): his name for the separation result the textbooks call Urysohn's lemma — in a normal space, two element-disjoint closed sets are separated by a continuous function equal to 0 on the one and 1 on the other. *Hauptsatz*: his name for the metrization result. A *normal space* (§3) is one in which any two mutually disjoint closed sets are separable by domains. A set is of *dimension 0* (§8) under his stated condition, otherwise of positive dimension. *Element-disjoint* renders *elementenfremd*; *Gebiet* is a domain, an open set.

The story

Pavel Samuilovich Urysohn was born on 3 February 1898 in Odessa, the son of an Odessa financier, and was educated at a private school in Moscow. In 1915 he entered Moscow University to study physics and published a first paper that

year, on Coolidge-tube radiation, under P. P. Lazarev; Luzin and Egorov drew him into pure mathematics. He graduated in 1919 and continued in doctoral study under Luzin, working on integral equations in 1919–21; the credential awarded in June 1921 was the habilitation, after which he became an assistant professor.

In the summer of 1921 Egorov set the Moscow school two problems — to define curve and surface intrinsically, by topology alone — and, on the folder's account, the definition of dimension came to Urysohn at Burkov near Moscow in late August 1921, and he told Aleksandrov of it at once. The dimension memoir followed, and with Aleksandrov, in 1923, the modern definition of compactness. On 17 August 1924, at Batz-sur-Mer, France, he began to write a new paper; that day he drowned while swimming in the Atlantic in rough seas, on a regular swim with Aleksandrov. He was twenty-six. After his death Brouwer and Aleksandrov took charge of his scientific estate.

The work and what survives

Two papers appeared posthumously in *Mathematische Annalen* 94 (1925). *Über die Mächtigkeit der zusammenhängenden Mengen* (pp. 262–295) is signed "Paul Urysohn †," dedicated "To my friend Paul Alexandroff," closed at Le Batz on 14 August 1924, and was received on 23 August 1924, six days after his death. *Zum Metrisationsproblem* (pp. 309–315) was received on 28 September 1924; only its first paragraph was edited by Urysohn himself, the rest after his death by Alexandroff, whose signed footnote states that oral communications had made him exactly acquainted with the whole plan and all details of the proofs.

What this project holds must be stated at its bounded scope. The folder holds *Zum Metrisationsproblem* complete in draft. Of the *Mächtigkeit* paper it holds §§1–11, §§24–25, and a short excerpt from §28; §§12–23, §§26–27, and the body of §28 are expressly untranslated. The draft awaits review by a working topologist and a reader of period German, including collation against the page images; uncertain readings are bracketed, never silently repaired.

Within that bound the chain can be followed: the cardinality theorems of §§9–11, the §25 Theorem, and the *Hauptsatz* — every normal space satisfying the second axiom of countability is metrizable — proved through $\rho(x, y) = \Sigma\, 2^{-n} \,|\, f_n(x) - f_n(y)|$, closing with an open question whether regularity alone might

suffice.

Its English fortunes

No English translation of either paper was found after research, though both theorems are restated in every textbook of topology.

That bound is the finding's whole strength. The theorems live in every textbook, proved afresh, but restatement is not the paper. In his own text the first result is simply "the Theorem," the second his *Hauptsatz*. No stronger claim is made.

What completing it would take

Completing the work, on the folder's own map, means translating the *Mächtigkeit* paper's missing sections: §§12–23, §§26–27, and the body of §28. It also means supplying the notes the folder defers to a NOTES.md that is not in the folder. What is already drafted must be collated against the page images, resolving the bracketed places, and the whole must pass the folder's gate — review by a working topologist and a reader of period German — before any publication. Alexandroff's posthumous editing must stay credited as his footnote credits it.

Sources

- Project folder: `~/workspace/rare-and-obscure-books/books/urysohn-zum-metrisationsproblem/urysohn-zum-metrisationsproblem.txt`
- PIN: `~/workspace/packages/boneyard-extension/research/PIN-urysohn.md`
- MacTutor History of Mathematics, Urysohn biography: https://mathshistory.st-andrews.ac.uk/Biographies/Urysohn/
- Springer, bibliography of metrization theory: https://link.springer.com/chapter/10.1007/BFb0064018
- MacTutor, earliest known uses of mathematical words: https://mathshistory.st-andrews.ac.uk/Miller/mathword/m/